MIL-P-83927B (USAF)
9 February 1981
Superseding
MIL-P-83927A(USAF)
8 April 1971

MILITARY SPECIFICATION

POWER SUPPLIES, GUIDED MISSILE LAUNCHERS, SINGLE TYPES PP-2401B/A, PP-2401C/A, PP-2401D/A, PP-4285/A, PP-4286/A, AND COMPOSITE TYPE PP-4506/A AND PS-10373C

This specification is approved for use by the AFLC Cataloging & Standardization Office, Department of the Air Force and is available for use by all Departments and Agencies of the Departmant of Defense.

1. SCOPE

1.1 General. This specification covens power supplies for the AIM-9 missle launcher. The power supply provides the missile with standby power, fixing power, motor drive power and amplification of audio signal.

1.2 Classification. The power supply contains low voltage and high voltage regulated supplies. For normal loads the low voltage supply is rated 25.2 VDC 1.5 amperes and the high voltage supply is rated 175 VDC 200 milliamperes.

NOTE: PP-4506 with PS-1037-3C forms one (two-section) power supply.

2. APPLICABLE DOCUMENTS

2.1 General. The following documents form a part of this specificat&m to the extent specified herein. The issues in effect on the date of invitation for bids or request for proposal shall apply unless particular issues are specified.

SPECIFICATIONS

FEDERAL	
PPP-B-601	Boxes, Wood, Cleted Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
MILITARY	
MIL-C-25	Capacitors, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metal Cases), General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODSAC, Federal Center, Battle Creek, MI. 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1440

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

MIL-T-27	Transformers and Inductors (Audio, Power and High-Power Pulse), General Specification for
MIL-P-116	Preservation - Packaging, Methods of
MIL-C-3432	Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra-Flexible, 300 and 600 Volts)
MIL-C-3965	Capacitors Fixed, Ekwkmlyhic (Nonsolid Electrolyte), Tantalum, General Specification for
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-C-5015	Connectors, Electrical, Circular Threaded General Specifications for
MIL-B-5087	Bonding, Electrical and Lightning Protection, for Aerospace Systems
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-R-5757	Relays, Electrical (for Electironie and Communication Type Equipment) General Specification for
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes
MIL-Q-9858	Quality Program Requirements
MIL-L-10547	Liner, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-R-22684	Resistors, Fixed, Film, Insulated~ General Specification for
MIL-R-39007	Resistors, Fixed, Wire-Wound (Power Type Established Reliability, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

	MIL-P-83927B (USAF)
MIL-R-39008	Resistors, Fixed, Composition (Insulated), Established Reliabl%ity, General Specification for
MIL-I-45208	Inspection System Requirements
MIL-C-45662	Calibration System Requirements

MILITARY STANDARDS

MIL-STD-129	Markings for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrinic Component Parts
MIL-STD-461	Electomagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-701	Lists of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-750	Test Methods for Semiconductor Devices
MIL-STD-781	Reliability Tests, Exponential distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-882	System Safety Program Requirements
MIL-STD-1520	Corrective Action and Disposition System for Nonconforming Material

PUBLICATIONS

ANA Bulletin 438	AGE Controls for Age-Sensitive Elastomeric Items

DRAWINGS

AIR FORCE (AF)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

AF 61D2434	Schematic PP-4285/A Power Supply
AF 61D2435	Outline Launcher Power Supply PP-4285/A
AF 63D22154	Outline - Launcher Power Supply PP-4286/A
AF 65F17180	Outline - Launcher Power Supply Type PP-4506/A with PS-1037-3C
AF 65D17181	Wiring Schematics Power Supply PP-4506/A & PS 1037-3C
AF 68D35618	Power Supply PP2401B/A
AF 69E32602	Schematic Diagram, Power Supply PP-2401B/A
AF 69E32973	Schematic Diagram - Power Supply PP-4286/A
AF 7132982	Power Supply PP-2401B/A
AF 7737988	Schematic Diagram - Power Supply PP-2401B/A
AF 7843128	Schematic Diagram - Power Supply PP-2401C/A & PP-2401D/A
AF 7843137	Outline - Launcher Power Supply PPP-2401C/A & PP-2401D/A

(Copies of specifications, standards, publications and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction and Production Tests. Power supplies covered by this specification are subject to the tests and provisions of Section 4.

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Parts and materials as approved by MIL-E-5400 shall be given first consideration.

b. When the characteristics of Standard MIL parts and materials are such that their use will not fulfill the requirements, substitute material and parts which most nearly meet or exceed the requirements of the respective standard parts or specifications shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

Nonstandard parks of an electrical or electronic nature as defined in MIL-E-5400 shall be approved by the AIM-9 engineering activity before the power supplies are submitted for preproduction testing. This request for approval must be submitted before the park is purchase or fabricated for these power supplies.

d. Parts and materials shall be used that conform to the environmental and electrical best requirements of MIL-STD-202.

e. Materials. F&cycled and reoovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requimments. Materials used shall be free from defects which would affect the Pergcxmanoe or maintianability of individual components of the overall assembly.

3.2.1 Transistors and Diodes. The choice and application of transistors and diodes used in the equipment shall be in accordance with MIL-E-4682, MIL-STD-701, MIL-STD-750 and MIL-S-19500. No vaccuum tubes nor plastic-encapsulated semiconductors shall be used.

3.2.2 Relays. Power supply relays shall conform to MIL-R-5757.

3.2.2.1 Relays K1, K2, K3 and K4. These relays shall have the following characteristics:

Coil Voltage	24 to 32 VDC at -65°C to +125°C
Coil Resistance	250 ohms minimum at 25°C
Pull-In Voltage	18 VDC maximum at -65°C to + 125°C
Drop-Out voltage	1.5 to 10 VDC at -65°C to +125°C
Ccntacts	Capable of carrying a minimum current of 5.0 amperes at 28 VDC with contact resistance not to exceed 0.05 ohm from 0.01 to 5.0 amperes at -65°C to +125°C Operating Time: 15 milliseconds or less including contact bounce at rated voltage at 25°C Release Time: 10 milliseconds or less at reated voltage at 25°C
Life	50,000 cycles comprising one make, one break of each set of contacts

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

3.2.2.2 Relay K5. This relay shall have the same specifications as 3.2.2.1 except that the contacts shall be capable of carrying a minimum current of 1.0 ampere at 28 VDC instead of 5.0 amperes.

3.2.3 Transformers. The transformers in this equipment shall be manufactured in accordance with the applicable paragraphs of MIL-T-27 and shall meet Group B inspection as applicable.

3.2.4 Cable and Connectors. The equipment shall use cable per MIL-E-5400 and shall use connectors in accordance with MIL-C-5015. Shielding shall be used as necessary to comply with MIL-STD-461. All connectors on the equipment shall be provided with reusable, protective, plastic caps. For PP-4285/A and PP-4506/A, the flexible molded cable shall meet the requirements of MIL-C-3432, Class MDC Flexibility F, with rubber molding per MIL-R-6855, Class 2, Graded 60 or equivalent.

3.2.5 Wiring and Configuration References. The following drawings provide electrical and mechanical envelope interface requirements:

TYPE	ELECTRICAL DRAWING	MECHANICAL DRAWING
PP-4285/A	AF 61D2434	AF 61D2435
PP-4286/A	AF 69E32973	AF 69D22154
PP-4506/A & PS-1037-3C	AF 65D17181	AF 65D17180
PP-2401B/A	See the following NOTE.	
PP-2401C/A & PP-2401D/A	AF 7843128	AF 7843137

NOTE: Power Supply PP2401B/A has two sets of reference drawings. Refer to AF68D35618 and AF 69E32602 for -10 and -30 configuration. Refer to AF 7132982 and AF 7737988 for -50 configuration.

3.2.6 Wire. Wire used in this equipment shall conform to the requirements of MIL-W-16878.

3.2.7 Resistors. Resistors in this equipment shall conform to the requirements of MIL-R-22684, MIL-R-39007 and MIL-R-39008.

3.2.8 Capacitors. Capacitors used in this equipment shall conform to the requirements of MIL-C-25 and MIL-C-3965.

3.3 Design and Construction.

3.3.1 Service Conditions. The equipment shall operate satisfactorily under any reasonable combination of service conditions specified in Specification MIL-E-5400 for Class 2 Equipments with the following exceptions and additions:

3.3.1.1 Equipment Position. Equipment shall meet all requirements of this specification regardless of orientation or position.

3.3.1.2 Shelf-Life. Shelf-life shall be a minimum of three years.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:53Z
Check the source to verify that this is the current version before use.

3.3.1.3 Under Voltage Frequency. The equipment shall not be damaged by operation below the voltage and/or frequency limits specified below and shall automatically resume normal operation upon return within those limits

3.3.1.4 Input Voltage Limits. The equipment shall operate as prescribed for power inputs of all voltage and frequency combinations of 102 to 124 Volts Root Mean Square, (VRMS) and 320 to 522 Hertz (Hz) from a power source otherwise conforming to MIL-STD-704.

3.3.1.5 Altitude. This power supply shall operate satisfactorily at pressures corresponding to 0 (zero) to 80,000 feet mean seal level (MSL).

3.3.1.6 Temperature. Continuous operation over a temperature range of -65 to +100 degrees Centigrade (°C) is required.

3.3.1.7 Overload Protection of Power Supply of Types PP-2401B/A and PP-2401D/A. Overload conditions are those voltages and/or currents which exceed the combined steady state and transient limits and are terminated by the action of protection devices. These power supplies shall be capable of withstanding steady state currents on the low votlage output greater then 3.0 amperes. Low voltage supply transient currents of 12 amperes for five milliseconds shall not activate built-in protection devices. These power supplies shall be capable of withstanding steady state currents on the high voltage output greater than 280 milliamperes. High voltage supply transient currents of .8 amperes for 10 milliseconds shall not activate built-in protection devices. Varying the load conditions between no load, transient load and normal load shall not activate built-in protection devices. The overload protection device shall activate for all steady state current loads of 3.5 amperes and/or 350 milliamperes or greater in the low voltage supply and high voltage supply respectively. When an overload condition exists in the low voltage and/or high voltage supply, the power shall be electrically disconnected from its load. Upon removal of the overload condition, the power supplies shall meet the requirements of 3.3.3.

3.3.2 Configuration. The physical layout and dimensions of the power supply shall be in accordance with applicable outline drawings of 2.1.

3.3.2.1 Weight. Maximum weights for the different power supplies are as follows:

a. PP-2401B/A, PP-2401C/A and PP-2401D/A -	9 pounds (lbs)
b. PP-4506/A and PS 1037-3C -	5.5 lbs
c. PP-4285/A and PP-4286/A -	11 lbs

3.3.3 Electrical Requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

3.3.3.1 High Voltage Regulated Supply. This supply shall convert 115 VRMS 400 Hz from the aircraft to furnish B+ power to the missile. The supply shall be regulated to produce 175 ± 10 VDC with an external load from 0 to 200 milliamperes and shall be regulated to 175 plus 10 minus 15 with an external load of 280 milliamperes. The specified output shall be obtained for input voltages of 102 to 124 VRMS at 320 Hz to 522 Hz through the temperature range of -65°C to +100°C. The maximum ripple on the output voltage shall not exceed 0.5 volts peak to peak between the temperature range of 25 to 75°C when under a 200 milliampere load. Ripple voltage shall not exceed 1.25 V peak to peak under any condition. All ripple voltages consisting of frequencies other than the fundamental and second harmonic of the input alternating current (AC) supply may be ignored. No damage shall result from a continuous short circuit between the high voltage regulated output and the low voltage regulated output with the high voltage power supply connected to its normal load.

3.3.3.2 Low Voltage Regulated Supply.

a. Low Voltage Supply of Types PP-4285/A and PP-4286/A. This supply shall convert the 115 volt MS 400 Hz from the aircraft to furnish power to the missile. The supply shall be regulated to 25.2 ± 1.2 VDC under loads ranging from 1.2 to 1.5 amperes. The specified output shall be obtained when the input voltage varies from 102 to 124 VRMS and from 320 to 522 Hz throughout the temperature range of -65°C to +100°C. The open circuit voltage shall be less than 29 VDC. The maximum ripple on the output voltage shall not exceed .15 volts peak to peak when under 1.2 to 1.5 ampere load. All ripple voltages consisting of frequencies from the fundamental and the second harmonic shall be included.

b. Low Voltage Supply of Types PP-2401B/A, PP-2401C/A and PP-2401D/A. This supply shall convert the 115 volt RMS 400 Hz from the aircraft to furnish power to the missile. The supply shall be regulated to 25.2 ± 1.2 VDC under loads ranging from 1.2 to 1.5 amperes. This specified output shall be obtained when the input voltage varies from 102 to 124 VRMS and from 320 to 522 hertz throughout the temperature range of -65°C to +100C. The open circuit voltage shall be less than 29 volts DC. The maximum ripple on the output voltage shall not exceed 0.030 VRMS when under 1.5 ampere load and 0.30 VRMS at 3.0 ampere load. Ripple measurements shall include all frequencies from the fundamental through the tenth harmonic of the 400 Hz input frequency. The power supply shall he regulated to greater than 21.0 VDC when subjected to 3.0 ampere load. The power supply shall be capable of delivering 3.0 amperes for 30 seconds without suffering damage.

c. Low Voltage Supply of Type PP-4506/A. This supply shall convert 93 VRMS 400 Hz from the PS-1037-3C supply to furnish power to the missile. The supply shall be regulated to 25.2 ± 1.2 VDC under loads ranging from 1.2 to 1.5 amperes. This specified output shall be obtained when the input voltage varies from 83.7 to 102.3 VRMS and from 320 to 522 Hz throughout the temperature range of -65°C to +100°C. The open circuit voltage shall be less than 29.0 VDC. The maximum ripple on the output shall not exceed .15 volts peak to peak when under 1.5 amperes load. All ripple voltages consisting of frequencies from the fundamental through the tenth harmonic of the 400 Hz

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

input frequency shall be included. No damage shall result from a continuous short circuit between the low voltage regulated output of this supply and the high voltage regulated output of the PS-1037-3C with the low voltage power supply connected to its normal load.

3.3.3.3 Firing Interlock Circuit.

a. Firing Interlock Circuit of Types PP-2401B/A, PP-2401C/A, PP-2401D/A and PP-4286/A.

(1) The resonance frequency of the tuned circuit shall be 6100 Hz ± 100 Hz.

(2) Relay K3 Shall close when an MS voltage of not less than 11 nor greater than 25 volts at the resonance frequency is applied between J4 - K and J4 - H.

(3) Relay K3 shall not pull in when an MS voltage of less than 80 volts at frequency of 4000 Hz + 50 Hz is applied between J4 - K and J4 - H.

(4) The maximum voltage applied between J4 - T and J4-H necessary to activate relay K3 shall be 18 volts DC.

(5) All of the above shall apply over the temperature range of -65°C to 100°C.

b. Firing Interlock Circuit of Type PP-4506/A.

(1) The resonance frequency of the tuned circuit shall be 6100 Hz ± 100 Hz.

(2) Relay K3 shall close when an RM3 voltage of not less than 11 volts nor greater than 25 volts at the resonance frequency is applied between P5-D and P5 - E.

(3) Relay K3 shall not pull in when an RMS voltage of less than 80 volts at a frequency of 4000 Hz ± 50 Hz is applied between P5 - D and P5-E.

(4) The maximum voltage applied between P5 - L and P5 - E necessary to activate relay K3 shall be 18 volts DC.

(5) All of the above shall apply over the temperature range of -65°C ± 100°C.

c. Firing Interlock Circuit of Type PP-4285/A.

(1) The resonance frequency of the tuned circuit shall be 6100 Hz + 100 Hz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

(2) Relay K3 shall close when an RMS voltage of not less than 11 volts nor greater than 25 volts at the resonance frequency is applied between J1 - K and J1 - H with J2 - P grounded.

(3) Relay K3 shall not pull in when an RMS voltage of less than 80_ volts at a frequency of 4000 Hz + 50 Hz is applied between J1 - K and J1 - H with J2 - P grounded.

(4) The maximum voltage applied between J1 - T (+) and J1 - H with J2 - P grounded necessary to activiat relay K3 shall be 18 volts DC.

(5) All of the above shall apply over the temperature range of -D650C to + 100°C.

3.3.3.4 Audio Amplifier.

a. Audio Amplifier of Types PP-2401B/A, PP-4285/A and PP-4286/A.

(1) The amplifier shall operate class "A" push-pull, and shall be so designed and constructed that a short circuit or open circuit across the output will not damage the amplifier. Input impedance shall be a minimum of 20,000 ohms.

(2) Gain characteristics shall be:

Input-Volts RMS at 840 and 1200 Hz	Output-Volts RMS into 600 - Ohms ± 1%, Non-Inductive Load
005	2.5 minimum
5	25 minimum
10 or greater	40 + 5

(3) The distortion at 5.5 VRMS input at 150, 840, and 2000 Hz shall not exceed 15%.

(4) With a 12,000 ohm (l%) resistor across the input of the audio amplifier, the random noise across a 600 ohm (1%) output load shall not exceed 0.15 VRMS.

b. Audio Amplifier of Type PS1037-3C.

(1) The amplifier shall operate class "A" push-pull, and shall be so designed and constructed that a short circuit or open circuit across the output will not damage the amplifier. Input impedance shall be a minimum of 20,000 ohms.

(2) Gain characteristics shall be:

Input-Volts RMS at 840 and 1200 Hz	Output-Volts RMS into 10 Ohms ± 1%, Non-Inductive Load
0.5	0.32 minimum
5	3.2 minimum
10 or greater	4.5 to 6.0

Source: https://assist.dla.mil Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

(3) The distortion at 5.5 VRMS input at 150, 840 and 2000 Hz shall not exceed 15%.

(4) With a 12,000 ohm (1%) resistor across the input of the audio amplifier, the random noise across a 10 ohm (1%) output load shall not exceed 6 millivolts RMS.

c. Audio Amplifier of Type PP-2401C/A and PP-2401D/A.

(1) The amplifier shall operate Class "A" and shall be so designed and constructed that a short circuit or open circuit across the output will not damage the amplifier. The voltage gain of the audio amplifier for all combinations of environmental conditions and input voltage extremes, shall be 1.6 + 5% for input voltage of 1.0 to 17 volts peak and shall be 1.6 plus 10% minus 20% for voltages of 17 to 24 volts peak. The maximum output of the amplifier for inputs of 24 to 30 volts peak shall be limited to 64 volts peak. This voltage gain shall apply for frequencies between 1100 and 1900 Hz. The amplifier gain may decrease by a total of 3DB from 1900 to 3000 Hz and from 1100 to 200 Hz. The amplifier gain shall decrease by at least 6DB per octave below 200 Hz and by at least 18 DB per octave above 3000 Hz. The conditions shall apply when the amplifier is operating into a 600 ± 1% ohm non-inductive load. The amplifier gain shall apply for a source impedance of 20K to 100K ohm providing an input voltage range of 0 to 30 volts peak.

(2) The distortion of 5.5 VRMS input at 150, 840 and 2000 Hz shall not exceed 15%.

(3) With a 12,000 ohm (1%) resistor across the input of the audio amplifier, the random noise across a 600 ohm (1%) output load shall not exceed 0.15 VRMS.

3.3.3.5 Electromagnetic Interference.

a. Electromagnetic Interference for Types PP-2401B/A, PP-4285/A, PP-4286/A, PP-4506/A and PS1037-3C. The generation of radio frequency interference (RFI) by the equipment and the susceptibility of the equipment to RFI shall be controlled within the limits of Class ID of MIL-STD-461 Notice 3.

b. Electromagnetic Interference for Types PP-2401C/A and PP-2401D/A. The generation of radio frequency interference (RFI) by the equipment and the susceptability of the equipment to RFI shall be controlled within the limits of Class ID of MIL-STD-461 Notice 3 with the following additions:

(1) The filed intensity level requirement of RS03 is increased to 20 volts/meter over the frequency range of 14KHz to 10 GHz.

(2) No degradation of performance shall occur when the system’s interconnecting and signal leads are bundled throughout their entire length with a wire carrying a 600 volt peak-to-peak relay switching transient.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:53Z
Check the source to verify that this is the current version before use.

(3) Spikes (transients of duration less than 50 microseconds) generated by the equipment shall not exceed +50% or -150% of the normal line voltage for DC power lines, + 50% of the normal RMS voltage for AC power lines or the limits specified-in the approved EMC qualification test procedure for other lines.

3.3.3.6 Bonding. The equipment shall be adequately bonded in accordance with MIL-B-5087.

3.3.3.7 Warm-Up. The equipment shall operate as prescribed within one minute after power applied.

3.3.4 Nomenclature and Nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-STD-130 and shall include:

a. Launcher Power Supply

b. Military Type Number

c. Air Force Stock Number

d. Part Number

e. Manufacturer's Identification

f. Serial Number

g. Date of Manufacture (month and year)

h. Contract Number

i. Input 28 VDC, 115 VAC, 400 Hz, Output Voltages

j. Full Load Input and Output Currents

3.3.4.1 Instruction Plate. A metal label shall be installed on each cable assembly indicating its part number. Another label shall be installed on or near each connector thereof to indicate its number.

NOTE: On type PP-4506/A and PS1037-3C a red warning metal label shall be located adjacent to the nameplate and inscribed as follows:

WARNING
UNIT R.F. GROUNDED
DO NOT SUBJECT TO DIELECTRIC
OR INSULATION RESISTANCE TESTS

3.4 Reliability. The specified mean-time-to-failure for this equipment shall be 1200 hours, its discrimination ratio 1.5:1 and the provisions of MIL-STD-781 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

3.4.1 Longevity. The life requirement for the equipment shall be 1500 hours.

3.5 Workmanship. The power supplies shall be assembled and finished in a thoroughly workman like manner. Particular attention shall be given to thoroughness of soldering, neatness and freedom of burrs and sharp edges. All power supplies procured under this specification shall be subject to Government rejection for poor workmanship.

3.6 Safety. Applicable portions of MIL-STD-882 are requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.9 General. The provisions of MIL-Q-98585 and MIL-STD-1520 shall apply except as otherwise specified herein. The supplier is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests set forth in this specification where such inspections or tests are deemed necessary to assure that supplies and service conform to prescribe requirements. Provisions of MIL-I-45208 and MIL-C-45662 shall apply to test requiren% used.

4.2 Classification of Tests. The inspection and testing of the power supplies shall be classified as follows:

a. Preproduction Testing (4.3)

b. Production testing (4.4)

4.2.3 Testing Procedures. The procedures used for conducting preproduction/qualification tests production acceptance tests and production reliability tests shall be prepared by the contractor and submitted to the preproduction/qualification tests production acceptance tests and produc- reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

4.2.2 Power and Load Conditions for Testing. The terminology discussed in this paragraph is used throughout Section 4 to simplify test instructions.

a. AC Power.

CONDITION	Type PP-2401B/A, PP-2401C/A PP-2401D/A, PP-4285/A, PP-4286/A, PS1037-3C	Type PP-4506/A
Normal voltage/ normal frequency	115 V RMS/400 Hz	93 V RMS/400 Hz

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

CONDITION	Type PP-2401B/A, PP-2401C/A PP-2401D/A, PP-4285/A, PP-4286/A, PS1037-3C	Type PP-4506/A
Normal voltage/ low frequency	115 V RMS/320 Hz	93 V RMS/320 Hz
Normal voltage/ high frequency	115 V RMS/522 Hz	93 V RMS/522 Hz
Low voltage/ normal frequency	102 V RMS/400 Hz	83.7 V RMS/400 Hz
Low voltage/ low frequency	102 V RMS/320 Hz	83.7 V RMS/320 Hz
Low voltage/ high frequency	102 V RMS/522 Hz	83.7 V RMS/522 Hz
High voltage/ normal frequency	124 V RMS/400 Hz	102.3 V RMS/400 Hz
High voltage/ low frequency	124 V RMS/320 Hz	102.3 V RMS/320 Hz
High voltage/ high frequency	124 V RMS/522 Hz	102.3 V RMS/522 Hz

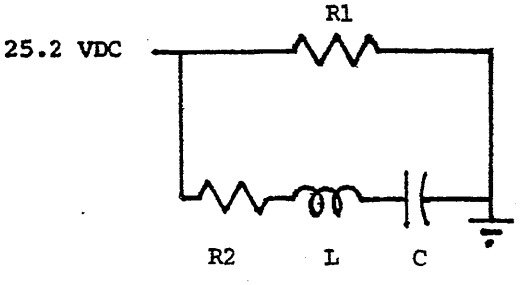
b. DC Power. Normal DC power shall be 28 VDC.

c. High Voltage Load. Normal high voltage supply load shall be 200 nilliamperes.

d. Low voltage Load. Normal low voltage load shall be 1.5 amperes.

e. High Voltage Ripple Circuit. Ripple measurements shall be taken under a 200 milliampere load.

f. Low Voltage Ripple Measurement. Ripple measurement shall be taken with the load circuit specified below:



1.5 Ampere Load	3 Ampere Load
R1 = 18 Ω ± 5%	R1 = 9 Ω ± 5%
R2 = 3 Ω ± 20%	R2 = 3 Ω ± 20 %
L = 260 μH ± 20%	L = 260 μH ± 20%
C = 130 μF ± 20%	C = 130 μF ± 20%

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

4.3 Preproduction Qualification Testing. The preproduction qualification tests are intended to provide assurance that the design and operation of the power supply will meet requirements. Successful completion of the test program does not relieve the contractor of the responsibility to furnish equipment that meets the performance requirements of this specification. These tests shall demonstrate that all requirements of 3.2, 3.3, 3.4 and 3.5 have been met. The preproduction power supplies shall be delivered to an independent test laboratory acceptable to the procuring activity for preproduction testing.

4.3.1 Scope of Tests. Power supplies submitted for testing shall be subjected to the testis following in The order and quantity indicated.

PREPRODUCTION TEST	PARAGRAPH	NUMBER OF SUPPLIES TESTED
Burn-in	4.3.1.1	All
Examination of product	4.3.1.2	All
Continuity test	4.3.1.3	All
Insulation resistance test	4.3.1.4	All
Performance test	4.3.1.5	All
Electromagnetic interference test	4.3.1.6	2
Environmental tests	4.3.1.7	3
Reliability test	4.3.1.8	6 (including 2 previously subjected to environmental tests)

4.3.1.1 Burn-In. The power supply shall be placed in a temperature test chamber, making such connections and instrumentation as is necessary to determine equipment operation and temperature stabilization. Initial temperature stabilization time at each temperature shall be ascertained by taking resistance measurements across a suitable component such as the audio input transformer or relay coil K2 every 15 minutes. When duplicate readings are obtained, stabilization has taken place. During temperature transition and stabilization, the power supply shall be nonoperating.

a. Set the chamber to +71°C. After stabilization, operate at normal power and load conditions for four hours.

b. Set the chamber to -65°C. After stabilization, operate at normal power and load conditions for four hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

c. The tests of 4.3.1.2, 4.3.1.3, 4.3.1.4 and 4.3.1.5 will check the operation of the power supply after the burn-in conditions.

4.3.1.2 Examination of Product. The power supply shall be examined to determine conformance to this specification with respect to materials, workmanship and design requirements.

4.3.1.3 Continuity Test. All components and wiring shall be tested to demonstrate compliance to the schematic drawing.

4.3.1.4 Insulation Resistance Test. A high potential of 500 VDC minimum shall be applied between each isolated circuit and other isolated circuits and between each isolated circuit and ground. All insulation resistances measured shall be 100 megohms minimum. Refer to schematic drawing for wiring information.

NOTE: Circuits containing diodes or other components that may be damaged by the high potential axe exempt from this test.

4.3.1.5 Performance Test.

4.3.1.5.1 High Voltage Regulated Supply.

a. High Voltage Supply of Types PP-2401B/As PP-2401C/A, PP-2401D/A, PP-4285/A, PP-4286/A and PS1037-3C. Apply each of the nine AC power conditions of 4.2.2 a.

(1) The output voltage shall be 175 ± 10 VDC with an open circuit load for each input condition.

(2) The ripple voltage shall not exceed 0.5 volts peak to peak with an open circuit load for each input condition. All ripple voltages consisting of frequencies other than the fundamental and second harmonic of the input voltage may be ignored.

(3) The output voltage shall be 175 DC with a 200 milliampere load for each input condition.

(4) The ripple voltage shall not exceed 0.5 volts peak to peak with a 200 milliampere load for each input condition.

b. High Voltage Supply of Types PP-2401B/A, PP-2401C/A, PP-2401D/A, PP-4285/A, PP-4286/A and PS1037-3C. Apply normal AC power.

(1) Apply a one minute load of 280 milliamperes. Verify that no damage results by measuring that the output voltage and ripple remain within. specification for the normal AC power normal lead condition,

(2) Apply a one minute short circuit between the high voltage and low voltage outputs with the high voltage output connected to its normal load. Verify that no damage results by measuring that the output voltage and ripple remain within specification for the normal AC power normal bad condition.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:53Z
Check the source to verify that this is the current version before use.

c. High Voltage Supply of Types PP-2401B/A and PP-2401D/A. Apply normal AC power.

(1) Apply a transient load of .8 amperes for 10 milliseconds and check to see that the high voltage supply does not electrically disconnect from its load. Regulated voltage shall be 175 ± 15 volts DC.

(2) Apply a variable load (no-load to full load to no load) and check to see that the regulated voltage is 175 ± 10 volts DC.

(3) Apply a load of 400 milliamperes and check that both high voltage and low voltage supplies disconnect electrically from their respective loads. Verify that no damage results by measuring that the output voltage and ripple remain within specification for the normal AC Power load condition.

4.3.1.5.2 Low Voltage Regulated Supply.

a. Low Voltage Supply of Types PP-4285/A, PP-4286/A and PP-4506/A. Apply each of the nine AC power conditions of 4.2.2 a.

(1) The output voltage shall not exceed 29 VDC with an open circuit load for each input condition.

(2) The ripple voltage shall not exceed 0.15 volts peak to peak with an own circuit load for each input condition. Ripple shall include frequencies from the fundamental through the tenth harmonic of the input voltage.

(3) The output voltage shall be 25.23 ± 2 VDC with a 1.5 ampere load for each input condition.

(4) The ripple voltage shall not exceed 0.15 volts peak to peak with the output load circuit of 4.2.2 for each input condition. Ripple shall include all frequencies from the fundamental through the tenth harmonic of the input voltage.

b. Low Voltage Supply of Types PP-2401B/A, PP-2401C/A and PP-2401D/A. Apply each of the nine AC power conditions of 4.2.2 a.

(1) The output voltage shall not exceed 29 VDC with an open circuit load.

(2) The output voltage shall be regulated to 25.21 1.2 VDC with loads ranging to 1.5 amperes.

(3) The ripple shall not exceed 0.030 VRMS with loads ranging to 1.5 amperes with the output load circuit of 4.2.2.f. Ripple shall include all frequencies from the fundamental through the tenth harmonic of the input voltage.

(4) The ripple shall not exceed 0.30 VRMS at 3.0 ampere load with the output load circuit of 4.2.2.f. Ripple shall include all frequencies from the-fundamental through the tenth harmonic of the input voltage.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:53Z
Check the source to verify that this is the current version before use.

c. Low Voltage Supply of Types PP-2401B/A and PP-2401D/A. Apply normal AC power.

(1) Apply a transient load of 12 amperes for 5 milliseconds and check to see that the low voltage supply does not electrically disconnect from its load. Regulated voltage shall be greater than 10 VDC.

(2) Apply a variable load (no-load to full load to no load) and check to see that the regulated voltage is between 29.0 VDC and 24.0 VDC.

(3) Apply a load of 3.5 amperes and check that both low voltage and high voltage supplies disconnect electrically from their respective loads. Verify that no damage results by measuring that the output voltage and ripple voltage remain within specification for normal AC power load conditions.

4.3.1.5.3 Relay Logic.

a. Relay Logic of Types PP-2401B/A, PP-2401C/A, PP-2401D/A, PP-4285/A, PP-4286/A, PP-4506/A and PS1037-3C. Apply 18 VDC power to each relay coil and verify that the relay has pulled in.

b. Firing Logic Circuit of Types PP-2401B/A, PP-2401C/A, PP-4285/A and PP-4286/A. Perform a simulated test firing and verify that DC. voltage is retained at the rocket motor and target detector pins after removal of the firing voltage input pulse. Pin connections are as follows:

TYPE	MASTER ARM	FIRING	ROCKET MOTOR	TARGET DETECTOR
PP-2401D/A	J1 Pin C	J1 Pin A	J2 Pin B	J2 Pin C
PP-2401B/A	J1 Pin C	J1 Pin A	J2 Pin B	J2 Pin C
PP-2401C/A	J1 Pin C	J1 Pin A	J2 Pin B	J2 Pin C
PP-4285/A	J2 Pin E	J2 Pin L	J2 Pin M	J2 Pin N
PP-2486/A	J1 Pin C	J1 Pin A	J2 Pin B	J2 Pin C
PP-4506/A	J1 Pin C to P5 Pin U	J1 Pin A	J2 Pin 1	J2 Pin 4

c. Firing Interlock Circuit of Types PP-2401B/A, PP-2401C/A, P7-2-SOID;A, PP-4285/A, PP-428G/A and PP-4506/A. Relay F-3 shall close when 25 VRMS 6100 Hz is applied to the resonant tone circuit and shall not close when 80 VRMS 4000 Hz is applied. Refer to 3.3.3.3 for applicable pin connections.

4.3.1.5.4 Audio Amplifier.

a. Audio Amplifier of Types PP-2401B/A, PP-2401C/A, PP-4285/A, PP-4286/A and PS1037-3C. Apply normal AC power and demonstrate that the output signal meets the specifications of 3.3.3.4 for the specified load and input voltage/frequency conditions.

4.3.1.6 Electromagnetic Interference Test. An electromagnetic interference test shall be conducted in conformance with 3.3.3.5. 28 VDC and normal AC power shall be applied and normal loads connected. Two power supplies shall be tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

4.3.1.7 Environmental Tests. Environmental tests shall be conducted in the order specified. Three power supplies shall be tested.

4.3.1.7.1 Temperature Altitude Test. The power supply shall be placed in the test chamber, making such connections and instrumentation as is necessary to ascertain equipment operation and temperature stabilization. Stabilization shall be ascertained by taking resistance measurements across a suitable component such as the audio input transformer or relay coil K2 every 15 minutes. When the duplicate readings are obtained, stabilization has taken place. During temperature transition and stabilization, the power supply shall be nonoperating.

a. Set the chamber to ambient temperature and altitude. After stabilization a performance test shall be conducted in accordance with 4.3.1.5.

b. Set the chamber to -65°C and sea level. After stabilization, a performance test shall be conducted in accordance with 4.3.1.5 except that only the three low voltage conditions shall be used wherever nine AC power conditions are specified.

c. With the chamber remaining at -65°C and sea level, apply low AC voltage 400 Hz and 28 VDC power for one hour with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three low voltage conditions shall be used wherever nine AC power conditions are specified.

d. Set the chamber to -65°C and 80,000 feet. After stabilization, a performance test shall be conducted in accordance with 4.3.1.5 except that only the three low voltage conditions shall be used wherever nine AC power conditions are specified.

e. Set the chamber to +100°C and sea level. After stabilization, apply high AC voltage 400 Hz and 28 VDC power for one hour with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.

f. Set the chamber to +71°C and sea level. After stabilization, apply high AC voltage 400 Hz and 28 VDC power for four hours with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.

g. Set the chamber to +71°C and 80,000 feet. After stabilization~ apply high AC voltage 400 Hz 28 VDC power for four hours with the output connected to its normal load. Then a performance test shall be conducted in aacordance with 4.3.1.5 except that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.

h. Set the chamber to +20°C and 80,000 feet. After stabilization, apply high AC voltage 400 Hz and 28 VDC power for four hours with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

b. Set the chamber to -65°C and sea level. After stabilization, a Performance test shall be conducted in accordance with 4.3.1.5 except that only the three low voltage conditions shall be used wherever nine AC power conditions are specified.

c. With the chamber remaining at -65°C and sea level, apply low AC voltage 400 Hz and 28 VW power for one hour with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three low voltage conditions shall be used wherever nine AC power conditions are specified.

d. Set the chamber to -65°C and 80,000 feet. After stabilization, a performance test shall be conducted in accordance with 4.3.1.5 except that only the three low voltage conditions shall be used wherever nine AC power conditions are specified.

e. Set the chamber to +100°C and sea level. After stabilization, apply high AC voltage 400 Hz and 28 VDC power for one hour with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.

f. Set the chamber to +71°C and sea level. After stabilization, apply high AC voltage 400 Hz and 28 VDC power for four hours with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.

g. Set the chamber to +71°C and 80,000 feet. After stabilization apply high AC Voltage 400 Hz 28 VDC power for four hours with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 exoept that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.

h. Set the chamber to +20°C and 80,000 feet. After stabilization apply high AC voltage 400 Hz and 28 VDC power for four hours with the output connected to its normal load. Then a performance test shall be conducted in accordance with 4.3.1.5 except that only the three high voltage conditions shall be used wherever nine AC power conditions are specified.
MIL-P-83927E (USAF)

4.3.1.7.3 Vibration Test. This test shall be conducted in conformance with MIL-STD-810B Notice 4, Method 514.1 Procedure I. Equipment category (b) curve H shall apply. During testing, normal AC power and 28 VDC shall be applied with the output connected to its normal load. Simulated test firing shall be conducted repeatedly throughout the testing. except that individual firing relays of the composite type supplies PP-4506/A and PS1037-3C shall be tested. Upon completion a performance test shall be conducted in accordance with 4.3.1.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

4.3.1.7.4 Mechanical Shock Test. This test shall be conducted in conformance with MIL-STD-810B Notice 4, Method 516.1 procedure I. Figure 516-2 with 30 g amplitude and 11 millisecond duration shall apply, During testing, the supply shall be nonoperating. Upon completion, a performance test shall be conducted in accordance with 4.3.1.5 except that only the normal voltage 400 hz condition shall be used wherever nine AC power conditions are specified.

4.3.1.7.5 Humidity Test. This test shall be conducted in conformance with MIL-STD-810B Notice 4, Method 507 Procedure I. During testing, protective covers shall be attached to the connectors of the power supply. Upon completion, a performance test shall be conducted in accordance with 4.3.1.5 except that only the normal voltage 400 Hz condition shall be used wherever nine AC power conditions are specified.

4.3.l.7.6 Salt Fog Test. This test shall be conducted in conformance with MIL-STD-810B Notice 4, Method 509 Procedure I except that the duration of salt spray shall be 168 hours instead of 48 hours. During testing, protective covers shall be attached to the connectors of the power supply. Upon completion of each of the salt fog and drying periods, the supply shall be examined for corrosion and subjected to a performance test in accordance with 4.3.1.5 except that only the normal voltage 400 Hz condition shall be used wherever nine AC power conditions are specified.

4.3.1.7.7 Acceleration Test. This test shall be conducted in accordance with MIL-STD-810B Notice 4, Method 513.1 procedure II, except that a 15 g test level shall apply for all orientations During testing, normal AC power and 28 VDC shall be applied with the output connected to its normal load. The operation of all relays shall be verified once under test conditions. Upon completion of testing, a performance test shall be conducted in accordance with 4.3.1.5.

4.3.1.8 Reliability Test. Six power supplies shall be subjected to the preproduction reliability test. At least two of the supplies shall have been subjected previously to the environmental tests of 4.3.1.7.

4.3.1.8.1 Testing Environment. Testing shall be performed in the following environment:

a. Temperature shall be -65°C to +71°C.

b. Altitude shall be seal level to 80,000 feet.

c. Relative humidity shall be uncontrolled to 95 percent.

d. Vibration shall be 2.2 g ± 10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points on the power supply. The duration of vibration shall be at least ten minutes at the end of each hour of equipment operating time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

e. The equipment shall be continuously environmentally cycled within a 15 hour period per following steps:

	ALTITUDE	TEMPERATURE
(1)	Sea Level	25 ± 5°C
(2)	80,000 ft	-65°C Maximum
(3)	Sea level	+71°C Minimum
(4)	Sea level	-65°C Maximum
(5)	80,000 ft	+71°C Maximum
(6)	Sea level	+71°C 95% relative humidity
(7)	Repeat above	Repeat above

4.3.1.8.2 Test Cycle and Stabilization. Reliability test time is the time during the environmental cycle in which input power is applied to the power supplies. Stabilization shall be ascertained by taking resistance measurements across a suitable component such as the audio input transformer or relay coil K2 every 15 minutes. When duplicate readings are obtained, stabilization has taken place. One stabilization times have been established for each condition, these times shall be used in subsequent cycles.

4.3.1.8.3 Electrical Conditions. During power-on conditions, normal AC power and 28 VDC shall be applied with the-output conneeted to its normal load. Simulated test firings shall be conducted repeatedly during each power-on period using a 19 minute on/1 minute off cycle. Counters shall monitor the relays operation. The individual firimg relays of the composite type supplies PP-4506/A and PS1037-3C shall be tested in lies of simulated test firings.

4.3.1.8.4 Reliability Assurance. During each 15-hour period; a performance test in accordance with 4.3.1.5 shall be conducted at one of the test conditions. Different conditions shall be specified for performance testing in subsequent periods so that all conditions are tested a similar number of times. It is the contractor’s respnsibility to verify by test results and data analysis that the power supply has mean-time-to-failure of 1200 hours: The provisions of Test Plan XXVI of MIL-STD-781 shall apply.

4.3.1.9 Failure. Failure is defined as the inability of the power supply to meet-the requirements of 4.3.1.1 through 4.3.1.8. The procuring activity shall be advised immediately of any failure. Continued testing after a failure is done at the manufacturer's own risk.

4.3.1.10 Preproduction Test Approval. Approval of the proproduction power supplies shall be made by the procuring activity upon satisfactory completion of all tests and review of the preproduction-test report. No production power supplies shall be delivered prior to the approval of the preproduction supplies. Prefabrication of production equipment prior to this approval is at the contractor’s own risk. The preproduction samples shall not be submitted as production equipment under the contract.

4.4 Production Testing. Power supplies covered by this specification shall be subjected to the production acceptance tests of 4.4.1 and the production reliability tests of 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:53Z
Check the source to verify that this is the current version before use.

4.4.1 Production Acceptance Tests. Each power supply shall be subjected to the tests following in the order specified.

4.4.1.1 Burn-In. Burn-in shall be conducted in accordance with 4.3.1.1.

4.4.1.2 Examination of Product. Examination of product shall be conducted in accordance with 4.3.1.2.

4.4.1.3 Continuity Test. Continuity test shall be conducted in accordance with 4.3.1.3.

4.4.1.4 Insulation Resistance Test. Insulation resistance test shall be conducted in accordance with 4.3.1.4.

4.4.1.5 Performance Test. Performance test shall be conducted in accordance with 4.3.1.5.

4.4.1.6 Failure. Failure is defined as the inability of the power supply to meet requirements of 4.4.1.1 through 4.4.1.5.

4.4.1.7 Production Acceptance Test Approval. Approval of each production power supply shall be made by the procuring activity or Government Inspector upon satisfactory completion of all tests. Failure of any of the test shall be cause for government rejection of the failed power supply.

4.4.2 Production Reliability Tests. Production reliability tests shall be conducted on each lot in conformance with MIL-STD-781 at a test facility acceptable to the procuring activity. Power supplies selected for production reliability tests shall first have passed the production acceptance tests.

4.4.2.1 Test Procedure. Testing shall be in accordance with 4.3.1.8.1 through 4.3.1.8.3. The procedures used for conducting preproduetion reliability tests, production acceptance tesks and production reliability acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval/disapproval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

4.4.2.2 Reliability Assurance. During each 15 hour period, a performance test in accordance with 4.3.1.5 shall be conducted at one of the test conditions. Different conditions shall be specified for performance testing in subsequent periods so that all conditions are tested a similar number of times. Test Plan V of MIL-STD-781 shall be used to determine the accept-reject criteria.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

4.4.2.3 Production Reliability Test Approval. Approval of the samples submitted shall be made by the procuring activity upon satisfactory completion of all tests. Failure of the samples to meet the acceptance criteria of Test Plan V shall be cause for rejection of the production lot represented by the samples.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Each Power supply shall be packaged in accordance with Method IIb of Specification MIL-P-116.

5.1.2 Level C. Each power supply shall be packaged to the degree required for protection under known favorable conditions during shipment, handling and limited tenure of storage in accordance with MIL-STD-794, such as:

a. Movement, storage and handling conditions as known to permit this level; or,

b. Shipments moving via other than ocean transportation are for immediate use at the first receiving activity.

5.2 Packing.

5.2.1 Level A. Power supplies packaged in accordance with 5.1.1 shall be packed containers conforming to PPP-B-601 Wood, Cleated-plywood, Overseas Type, PPP-B-621 Wood, Nailed and lock-corner, Class 2, PPP-B-636 Fiberboard, Weather Resistant or PPP-B-640 Fiberboard, Corrugated Triple-wall Class 2. Boxes shall be sealed, closed and strapped for overseas shipment in accordance with applicable box specification and appendix thereto. Wood or cleated boxes shall be lined with sealed wategprd case liner conforming to MIL-L-10547, except that When the exterior container of Method IIb packages has strength and water resistance as specified in PPP-13-636, a case liner will not be required. The gross weight of wood, wood-cleated or triple wall corrugated boxes shall not exceed 200 pounds, fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.2.2 Level C. Power supplies shall be packed in containers in a manner to insureance and safe delivery at destination. Containers shall comply with the consolidated freight classification rules or other regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

5.4 Cure Dates. The contractor shall comply with the requirements of ANA Bulletin 438 as applicable.

6. NOTES.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

6.1 Intended Use. Power supplies covered by this specification are intended to supply standby power and fixing power to the AIM-9 missiles and amplify signals from the missile.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Levels of packaging and packing required (see 5.1 and 5.2).

c. Serial numbers applicable to contract quantities of test sets.

6.3 Identification. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensiveness of the damages.

Custodians:

Air Force - 99

Preparing Activity

Air Force - 99

Air Force Project No.

1440-F0558

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-P-83927B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:53Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

HQ AFLC CASO/LODSAC
FEDERAL CENTER
BATTLE CREEK, MI 49016

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:53Z
Check the source to verify that this is the current version before use.